

Chapter 1: Construction Management Framework [3 hours]

- **1.1 Construction Landmarks**

➤ *No direct questions asked in the past.*

- **1.2 Scope of Construction Management**

1. Discuss on scope of construction management. [2] (2075 Bhadra)
2. Discuss on scope of construction management. [Part of 5] (2070 Bhadra)
3. Discuss on scope of construction management. [Part of 6] (2073 Magh)
4. Elaborate the scope of construction management. [Part of 5] (2074 Magh)

- **1.3 Construction Project Characteristics**

1. What are the characteristics of Construction Project? Explain. [4] (2071 Bhadra)
2. The construction projects faces delay and cost over-majorly due to management failures. Do you agree with this statement? Explain in brief. Discuss on different construction environment which effects on construction project. [6] (2079 Shrawan)

- **1.4 Construction Project Life Cycle Phases**

1. Write the various phases of project life cycle. [Part of 10] (2072 Ashwin)
2. Discuss on project life cycle and phases. [Part of 6] (2073 Magh)
3. Discuss on project life cycle phases. [Part of 5] (2070 Bhadra)
4. Explain different phases on construction project life cycle with characteristics of each phase. [Part of 5] (2074 Magh)
5. What is construction project life cycle? Explain its phases with suitable diagrams. Mention which phase of project life cycle in Mega projects is more problematic in Nepal. [1+4+1] (2080 Chaitra)

- **1.5 Construction Project Management**

1. Define construction project management. [1] (2072 Magh)
2. Define construction management. Why construction team needs knowledge about construction management? [1+4] (2077 Chaitra)
3. Write down the function of construction project management in context of Nepal as Nepal needed more infrastructure planning and development. [4] (2073 Bhadra)

4. How could "Construction Management" course helps in addressing the problems related to delay, poor quality in delivery and exceeding the budgeted cost of a project? [4] (2079 Jestha)

- **1.6 Relation between Client, Consultant, and Contractor**

1. Define client. [Part of 4] (2074 Bhadra)
2. Discuss the relationship between client, consultant and contractor for proper execution of construction project. [5] (2078 Chaitra)
3. Discuss the relation between Client, Consultant and Contractor in a construction project. [Part of 10] (2072 Ashwin)
4. Discuss the relationship between Client, Consultant and Contractor. [4] (2071 Magh)
5. Explain about the relationship between client, consultant and contractor to perform the project effectively. [3] (2075 Bhadra)
6. Explain the relationship between owner, contractor and consultants. [4] (2069 Bhadra (Old Syllabus))
7. What relation between construction team (client, consultant and contractor) should be in construction project in order to complete project within schedule, time and cost? [5] (2078 Kartik)
8. Explain what relation between construction team (client, consultant and contractor) should be in construction project in order to complete project within schedule time and cost. [4] (2072 Magh)
9. Discuss most important two roles each of client, consultant and contractor. [4] (2074 Bhadra)
10. Define Consultant. Explain the role of Consultant in maintaining the quality of Construction works. [1+3] (2070 Magh)
11. Define consultant. Explain the role of consultant in maintaining the quality of construction works. [4] (2081 Shrawan)
12. List out various activities to be carried in the construction phase from the Employer's perspective. Describe the relationship between Client, Contractor and Consultants in three actors system. [2+3] (2079 Chaitra)

Chapter 2: Construction Planning and Scheduling [5 hours]

- **2.1 Construction Planning – Introduction**

1. Define planning. [1] (2071 Magh)
2. Define planning. [Part of 5] (2080 Chaitra)
3. Explain necessity of construction planning. Write in details about construction planning in a construction project to be carried out by a contractor in context of Nepalese Construction. [2+6] (2072 Magh)

- **2.2 Steps and Stages of Planning**

1. Explain the importances and steps of construction planning and scheduling. [Part of 10] (2072 Ashwin)
2. Discuss on Time cost Trade-off with examples and steps in planning. [5] (2070 Bhadra)

- **2.3 Planning by Contractor and Clients in Different Stages**

1. What are the different steps of planning carried out by a client in pre-tender stage of any construction project? Enlist and elaborate in bullet. [3] (2081 Shrawan)
2. Discuss the role of a client during pre tender stage. [Part of 4] (2074 Bhadra)
3. Write short notes on: Pre-tender stage planning by client. [3] (2079 Jestha)
4. State in details about planning activities carried out in implementation stage by the client of a project. [4] (2077 Chaitra)
5. What is the time frame for pre-tender stage for the contractor? [1] (2079 Chaitra)
6. Explain pre-tender stage or contract stage planning by contractor. [Part of 5] (2080 Chaitra)
7. Explain pretender Stage Planning by contractor in Detail. [4] (2070 Magh)
8. Explain pre-tender stage or contract stage planning by contractor. [5] (2071 Magh)

- **2.4 Preparing Schedule**

1. Why do you prefer Bar Chart as planning tool? [2] (2070 Magh)
2. Among CPM, PERT and bar chart which planning tool do you prefer for building construction? Explain. [8] (2069 Bhadra (Old Syllabus))

• 2.5 Time Cost Trade Off

❖ Theory

1. What do you mean by time-cost trade off? Explain the use of time cost trade off with example. [6] (2074 Bhadra)
2. Explain time cost trade off with example. [6] (2071 Bhadra)
3. What is time-cost trade off? Explain its importance for fixing duration of a construction work. [5] (2074 Magh,)
4. Explain the concept of time cost trade off in determining duration of contract. [5] (2072 Ashwin)
5. Write short notes on: Time cost trade off. [2] (2072 Magh)

❖ Numericals

1. Find out optimum project duration and minimum project cost using time cost trade off analysis. The indirect cost of the project is Rs.300/day. [8] (2081 Shrawan)

Activity	Successor	Time (Days)		Cost (Rs.)	
		Normal	Crash	Normal	Crash
A	C	5	4	1000	1200
B	D,E	7	5	2100	2300
C	F	3	2	1500	1800
D	F	2	2	1300	1300
E	G	6	4	1200	1400
F	-	5	4	2000	2500
G	-	3	2	1500	1700

2. Find the optimum project duration, minimum project duration and their corresponding project completion cost based on the given information. Consider all the rupees in Million NPR and durations in months. Time-dependent variable cost is Rs 210 per unit duration. [8] (2080 Chaitra)

Activity	Predecessor	Normal Duration	Crash Duration	Normal Cost (RS)	Crash Cost (RS)
A	-	12	8	600	1080
B	-	11	9	1500	1580
C	A, B	12	9	1000	1030
D	A, B	14	10	500	900
E	A	9	9	1500	1500

3. The details of project are shown below. If the indirect cost per week is Rs 450, find the optimal crashed result of the project network. [8] (2079 Chaitra)

Activity	Predecessor	Time (week)		Cost (Rs)	
		Normal	Crash	Normal	Crash
A	-	7	4	1800	2150
B	-	9	7	3500	3850
C	B	5	4	2500	2675
D	A	8	5	4000	4275
E	C	9	8	3000	3375
F	B	11	11	3000	-

4. From the given data in table below.

Activity	Predecessor	Normal Duration (days)	Crash Duration (days)	Normal cost (Rs.)	Crash Cost (Rs.)
A	-	5	2	5000	6200
B	A	4	3	6000	7000
C	A	3	2	5000	6500
D	A	6	4	4000	4400
E	B	5	3	15000	18000
F	D	4	2	3000	4800
G	C,E,F	6	3	8000	10400

Calculate minimum cost and minimum duration schedule using time cost trade off analysis.

Indirect cost is Rs.2000/day. [8] (2079 Shrawan)

5. Perform time-cost trade-off based on given information regarding a project comprising five different activities. Also, draw the time-cost trade-off chart. Take time dependent cost per week of Rs 1.6 thousand. [8] (2079 Jestha)

Activity	Predecessors	Time (Week)		Cost (Thousand Rs.)	
		Normal	Crash	Normal	Crash
A	-	26	18	4	20
B	-	28	22	5	20
C	A	10	5	6	11
D	A	4	2	8	11
E	B,D	12	6	3	14

6. Consider the data of a project as shown in table below:

Activity	Immediate Predecessor	Normal Time (Week)	Normal Cost (Rs.)	Crash Duration (Week)	Crash Cost (Rs.)
P	-	7	2500	4	2200
Q	-	9	4500	7	4200
R	P	5	3500	4	3000
S	P	8	5500	7	5100
T	Q	9	3000	8	2500
U	Q	12	5500	12	-
V	S, T	4	1500	2	1800

If the indirect cost per week is Rs. 2500, find the optimal crashed project completion time and corresponding minimum cost. [7] (2078 Chaitra)

7. Calculate the minimum cost, optimum duration using time cost trade off analysis for the following project schedule consisting of seven activities. Also find the possible minimum time to complete the project. Indirect cost of the project is Rs.225 per day. [8] (2078 Kartik)

S.N.	Project Activity	Immediate Predecessor	Duration (days)		Cost (NPR)	
			Normal Duration	Crash Duration	Normal Cost	Crash Cost
1	A	-	6	4	1400	1900
2	B	-	8	5	2000	2800
3	C	A	4	2	1100	1500
4	D	A	3	2	800	1400
5	E	B, C	6	3	900	1600
6	F	D	10	6	2500	3500
7	G	E	3	2	500	800

8. Consider the data of a project as shown in table below.

Activity	Immediate Predecessor	Normal Duration (Week)	Normal Cost (Rs.)	Crash Duration (Week)	Crash Cost (Rs.)
A	---	8	2000	5	2300
B	---	10	4000	8	4300
C	A	6	3000	5	3125
D	A	9	5000	6	5225
E	B	10	2500	9	2700
F	B	13	5000	13	---
G	D,E	5	1000	3	1700

If the indirect cost per week is Rs. 2000, find the optimal crashed project completion time and corresponding minimum cost. [16] (2076 Bhadra)

9. The details of a project are shown below. If the indirect cost per week is Rs. 400, find the lowest cost schedule of the project. [7] (2075 Bhadra)

Activity	Predecessor (s)	Time (weeks)		Cost (Rs)	
		Normal	Crash	Normal	Crash
A	-	8	5	2000	2200
B	-	10	8	3800	4000
C	B	6	5	2700	2600
D	A	9	6	4100	4300
E	C	10	9	3100	3350
F	B	12	12	3100	3100

10. The details of a project are shown below. If the indirect cost per week is Rs. 300, find the optimal crashed result of the project network. [6] (2073 Bhadra)

Activity	immediate Predecessor(s)	Time (Weeks)		Cost (Rs.)	
		Normal	Crash	Normal	Crash
A	-	7	4	1800	2100
B	-	9	7	3500	3800
C	B	5	4	2500	2625
D	A	8	5	4000	4225
E	C	9	8	3000	3325
F	B	11	11	3000	-

11. Consider the data of a project as shown in table below

Activity	Immediate Predecessor	Normal time (Week)	Normal Cost (Rs.)	Crash Duration (Week)	Crash Cost (Rs.)
A	-	8	2000	5	2300
B	-	10	4000	8	4300
C	A	6	3000	5	3125
D	A	9	5000	6	5225
E	B	10	2500	9	2700
F	B	13	5000	13	-
G	D, E	5	1000	3	1700

If the indirect cost per week is Rs. 3000, find the optimal crashed project completion time and corresponding minimum cost. [6] (2073 Magh)

12. Calculate Minimum Cost and Minimum Duration Schedule using time cost Trade-off analysis of the project with given detail. Indirect cost is Rs. 1000 per day [6] (2070 Magh)

Activity	Predecessor	Normal Duration	Crash Duration	Normal Cost	Crash Cost
A	-	5	2	5000	6200
B	A	4	3	6000	7000
C	A	3	2	5000	6500
D	A	6	4	4000	4400
E	B	5	3	15000	18000
F	D	4	2	3000	4800
G	C,E, F	6	3	8000	10400

13. For the activities lying on the critical path of a network, the normal duration and the crash duration along with their respective direct cost are given in the table.

Activity	Normal duration	Normal cost (Rs)	Crash duration	Crash costs (Rs)
1-2	2 days	50	1 day	75
2-3	4 days	160	3 days	225
3-5	9 days	270	6 days	350
5-6	5 days	100	4 days	250

The overhead indirect cost is Rs 30/- per day. Crashing is possible for 1 day for each of the activities 1-2, 2-3, and 5-6 and 3 days for activity 3-5. Find the lowest cost schedule by crashing step by step assuming no fresh critical path is developed on crashing a project. [8] (2069 Bhadra (Old Syllabus))

Chapter 3: Planning Construction Material [5 hours]

• 3.1 ABC Classification of Construction Materials

1. Explain ABC Classification of Construction Materials. [Part of 5] (2079 Shrawan)
2. Explain ABC classification of Construction material. [Part of 6] (2073 Magh)
3. Explain about ABC classification of construction material graphically. [3] (2073 Bhadra)
4. Define A, B and C category of materials. [3] (2070 Magh)
5. What do you mean by ABC classification of materials? Discuss importance of material management. [5] (2074 Bhadra)
6. What are the key benefits of ABC classification of materials? Explain its basis of ABC classification of materials. [2+2] (2080 Chaitra)
7. Which classification basis for materials would be beneficial for planning the construction materials and why? [2] (2079 Jestha)
8. Write short notes on: ABC Classification of materials. [4] (2070 Bhadra)

9. Write short notes on: ABC classification of construction materials. [2] (2072 Magh)
10. Write short notes on: ABC classification in material management. [5] (2074 Magh)
11. Consider a construction site acquired different materials given below, based on the data classify the materials in A, B and C classes. What are the significances of this classification? [4+2] (2077 Chaitra)

Materials	Quantity	Unit	Rate per unit
steel re-bar	15	ton	110000
cement	600	bag	750
tiles	1500	sqft	350
brick	9000	no	12
marbles	450	sqft	250
aggregate	50	cum	1400
sand	25	cum	1200

• 3.2 Material Wastage Standards

1. What is material wastage standard? [Part of 6] (2081 Shrawan)
2. What is material wastage standard? [Part of 6] (2071 Bhadra)
3. Explain importance of materials wastage standards. What are the different causes of materials wastage in construction project? Explain. [2+3] (2079 Chaitra)
4. What are the reasons of excessive material wastage during usage? Explain its preventive measures. [3+3] (2070 Magh)

• 3.3 Material Provisioning Process

1. Explain about the material provisioning process. [4] (2078 Chaitra)
2. Explain material provisioning process in detail. [4] (2071 Magh)
3. If you are appointed as project manager for a building project in rural area, write down the steps which you will be following for procurement of the materials, based on material provisioning process? [4] (2080 Chaitra)
4. Write short notes on: Process of material procurement. [4] (2069 Bhadra (Old Syllabus))

• 3.4 Material Inventory Basics

❖ Theory

1. Define EOQ. [1] (2078 Chaitra)
2. Define economic order quantity. [Part of 6] (2071 Bhadra)

3. Explain the terms: inventory, inventory cost, EOQ, cycle time, lead time, re-order level, good in transit, safety stock. [5] (2074 Magh)
4. What is Economic Order Quantity (EOQ)? Derive its expression. [4] (2081 Shrawan)
5. Write short notes on: Material inventory. [4] (2070 Bhadra)

❖ **Numericals**

1. A construction site of residential building with large floor area requires 6000 bags of cement in a year where a material supplier promised to supply within the five working days after placing the order. If site requires storage area separately which costs 15% storing charge per unit per year at that locality? Construction work will be carried for 200 days in a year and site engineer is planning for ordering the quantity of cement at a time by considering ordering cost per order of Rs 25000 charged by material supplier. What quantity can be suggested as EOQ. Is there any goods in transit (GiT)? If yes, calculate GiT too. What would be the re-order level, if site engineer wants to minimum 190 bags of cement as minimum safety stock? [6] (2079 Jestha)
2. Calculate EOQ, Maximum Stock level, minimum, average stock level, and ROP/ROL if annual cement required for a construction project is 20000 bags; ordering cost is Rs, 10,000; Carrying cost is 5%; Cement rate is Rs. 900 per bags; Safety stock is 10 days; lead time is 5 days. Also calculate the total inventory cost. [6] (2078 Kartik)
3. Calculate EOQ and reorder point if rebar required is 1000 tonnes; Ordering Cost is Rs.10,000; Carrying Cost is 15%, rebar rate is Rs.90,000 per tonne; Safety stock is 50 tonnes; lead time is 4 day and consumption per day is 50 tonnes. [Part of 6] (2071 Bhadra)
4. Calculate Economic Order Quantity of Cement with following details; Total Quantity = 12000 bags, $C_i=15\%$, $C_o=10000$, $P = 750$ Rs. per bag. [3] (2070 Magh)

• **3.5 Inventory Planning Process**

1. Discuss Inventory Planning Process for repetitive materials. [Part of 5] (2079 Shrawan)
2. Discuss inventory planning process for repetitive material. [Part of 6] (2073 Magh)
3. How inventory of construction materials can be planned for the repeatedly used items? Explain with suitable diagram. [2] (2077 Chaitra)

• **3.6 Application of Value Engineering in the Procurement of Materials**

1. What do you mean by Value Engineering? [2] (2078 Chaitra)
2. Write the importance of value engineering. [2] (2071 Magh)

3. How the value of a construction materials can be analysed on the basis of value engineering? Discuss it. [3] (2073 Bhadra)
 4. Describe the application of value engineering in procurement of material. [Part of 6] (2081 Shrawan)
 5. Describe the application of value engineering in procurement of material. [Part of 6] (2071 Bhadra)
 6. Write the application of value engineering in the procurement of construction materials. [Part of 10] (2072 Ashwin)
 7. Write short notes on: Value engineering. [3] (2075 Bhadra)
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Chapter 4: Familiarization with Construction Equipments [7 hours]

- **4.1 Advantages and Disadvantages of Using Equipments**

1. Enlist advantages and disadvantages of using equipment in construction. [2] (2079 Jestha, Q2b)
2. What are the prominent advantages and disadvantages of using construction equipments? Explain in short. [3] (2075 Bhadra)
3. Write advantages and disadvantages of using equipment in construction work. [5] (2074 Bhadra)
4. Explain the merits and demerits of using equipments. [Part of 6] (2073 Magh)
5. How construction equipment is considered disadvantages in construction project? Explain. [2] (2077 Chaitra)
6. What are the disadvantages of using equipment in Construction sites? [2] (2070 Magh)

- **4.2 Equipments for Excavation, Transporting and Compaction, Aggregate Production and Handling; Concrete Construction; Cranes for Lifting**

- **❖ Equipments for Excavation**

1. What are various equipment used for excavation works? List them and explain any two with proper sketches. [6] (2079 Shrawan)
2. Explain equipment for excavation. [Part of 10] (2072 Ashwin)

3. List out equipments required in excavation. Explain the function of excavator. [2+4] (2071 Magh)
4. Draw a line diagram of front-end loader and dragline and explain their functions in details. [4] (2079 Jestha)
5. Explain the equipment with the help of neat sketches: Backhoe [1.5] (2075 Bhadra)
6. Write short notes on grader. [4] (2072 Magh)
7. Write short notes on grader and dragline. [5] (2074 Bhadra)
8. Write short notes on : Face shovel and Front end loader [5] (2076 Bhadra)

❖ Equipments for Transportation and Compaction

1. Explain any two earth moving equipments in details with their suitable uses. Make sketches of them. [5] (2074 Magh,)
2. Explain equipment for compaction. [Part of 10] (2072 Ashwin)
3. Define earth compaction equipment. List out it and explain each with neat sketches. [6] (2078 Kartik)
4. List the equipment used in soil compaction with their suitability as per nature of soil. [8] (2081 Shrawan)
5. List the equipment used in compaction and explain any two of them with a neat sketch. [2] (2079 Chaitra)
6. Enlist the construction equipment used in earthwork and explain any two of them with neat sketches. [2+4] (2077 Chaitra)
7. Explain the key characteristics of following construction equipment with the suitability based on their functions: [6] (2080 Chaitra)

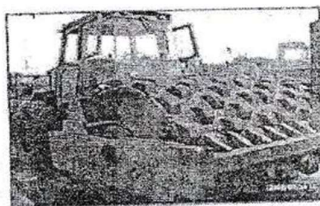


Figure (a)

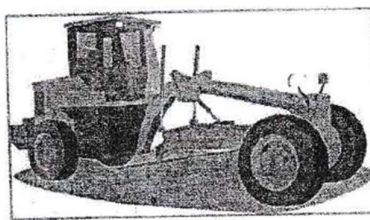


Figure (b)

❖ **Aggregate Production and Handling Equipments**

➤ *No direct questions asked in the past.*

❖ **Concrete Construction Equipments**

1. List the different type of equipments for concrete construction with their function. Explain briefly. [6] (2073 Magh)
2. Explain the function of concrete batching and mixing plant with neat sketches. [6] (2079 Chaitra)
3. Discuss Batching and Mixing plant. [6] (2071 Bhadra)
4. Explain the equipment with the help of neat sketches: Concrete mixer. [1.5] (2075 Bhadra)
5. Write short notes on: Concrete Batching Plants. [4] (2070 Bhadra)

❖ **Crane for Lifting**

1. List the different types of cranes and discuss in brief with sketches. [5] (2076 Bhadra)
2. What are the factors to be taken into consideration while selecting cranes for lifting of materials? [4] (2072 Magh)
3. Write short notes on: Tower Cranes. [4] (2070 Bhadra)

❖ **Tunnel Construction**

1. Discuss about the different equipments for tunnel construction. [8] (2069 Bhadra (Old Syllabus))
2. Discuss the methods of tunnel excavation. [6] (2071 Bhadra)
3. Explain the steps to be followed in sequence while carrying out tunneling work by using conventional method. Introduce TBM and its advantage in tunnel construction. [7] (2078 Chaitra)
4. Write short notes on equipment in tunnel construction. [4] (2072 Magh)
5. Write short notes on: Tunnel excavation by Blasting. [4] (2070 Bhadra)
6. Write short notes on: TBM [5] (2076 Bhadra)

❖ **Equipment for Hydraulic Construction**

1. List out construction equipment used in hydropower and explain any two of them with sketches. [3+5] (2070 Magh,)

❖ **Equipment for Highway and Pavement Construction**

1. Enlist the equipment for highway construction and explain any two of them with sketches. [2+6] (2069 Bhadra (Old Syllabus))
2. Write short notes on equipment used for highway and pavement construction. [4] (2073 Bhadra)
3. List out equipments used in road construction. Explain the function of paver. [2+4] (2071 Magh)
4. Write short notes on: Road pavers. [4] (2070 Bhadra)

• **4.3 Selection of Appropriate Equipment**

1. Explain the method of selection of appropriate equipment. [2] (2079 Chaitra)
 2. What are considerable factors for selection of a particular equipment? [2] (2079 Jestha)
 3. What are the factors that govern the selection of particular equipment in construction project? Explain. [5] (2074 Magh)
 4. Write the factors to be considered in the selection of appropriate construction equipment. [Part of 10] (2072 Ashwin)
 5. What are the points that should be consider while selecting appropriate equipment? [6] (2073 Magh)
 6. Discuss why economic and commercial considerations are vital factors affecting in selection of proper construction equipment with examples. [6] (2078 Kartik)
 7. Write down the factors to be considered for deciding the extent of mechanisation of construction project. [4] (2073 Bhadra)
 8. Write short notes on: Purchase or lease of equipment. [4] (2069 Bhadra (Old Syllabus))
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Chapter 5: Contract Management

• **5.1 Method of Work Execution**

1. Enlist different methods of execution of work guided by PPR? [2] (2079 Jestha)

• **5.2 Contract**

1. What is contract and what are its essential elements? Explain how it is important in any large scale construction project? (2078 Kartik) [4+2]

2. Describe Contract, essential elements of contract and types of contract in detail. [1+2+4] (2076 Bhadra)
3. List out different types of contract used in construction project. [Part of 6] (2079 Chaitra)
4. Name types of contract. [Part of 6] (2073 Magh)
5. Name contract types and reimbursable contract. [4] (2073 Bhadra)
6. Name the types of Contract. Explain the item rate contract. [6] (2075 Bhadra)
7. Classify contract based on payment. Explain lump sum contract and cost plus contract. [2+2+4] (2071 Bhadra)
8. Write short notes on: Unit price contract and its disadvantages. [3] (2077 Chaitra)
9. Explain different types of contract based on technical and administrative responsibility. [5] (2078 Chaitra)
10. Explain EPC contract in brief. [2] (2080 Chaitra)
11. Make differences between contract and agreement. Discuss turnkey and BOOT contract briefly. [5] (2074 Magh)
12. Name types of contract and discuss the use of BOOT contracts in the context of Nepal. [6] (2074 Bhadra)
13. Explain BOOT contract and Turn-Key contract. [4+4] (2079 Shrawan)
14. Explain with suitable examples: EPC contract, Turn key contract, BOO contract and BTO contract. [4] (2079 Jestha)
15. What is BOOT Contract? Explain its importance in Developing Countries like Nepal. [2+4] (2070 Magh)
16. Write short notes on: BOOT. [4] (2073 Bhadra)
17. Write short notes on: Types of contract. [4] (2070 Bhadra)
18. Explain about the different types of contract practicable in our context. [3] (2075 Bhadra)

- **5.3 Tendering (Bidding) Process**

1. What is Tender Notice? [4] (2079 Shrawan)
2. Define Tender Notice and write the essential information that should be included in a Tender Notice. [5] (2072 Ashwin)
3. What is tender notice? Enlist the task for a contractor during submission of tender. [5] (2074 Magh)
4. Write down the activities to be carried out before inviting tender. [3] (2075 Bhadra)
5. Write down the activities to be carried out as preparation before inviting tendering. [4] (2073 Bhadra)

6. What is tender document? [Part of 5] (2074 Magh)
7. Explain the content of tender document. [Part of 6] (2073 Magh)
8. Differentiate between tender document and contract document. [Part of 6] (2079 Chaitra)
9. Define bid security in contract and explain its use. Mention what are the essential informations to be provided for advertising tender notice in media. [4+4] (2072 Magh)
10. Explain Bid security in brief. [2] (2080 Chaitra)
11. Write short notes on Tendering Process. [5] (2079 Shrawan)
12. Prepare a sample Tender Notice for procurement of works for NCB contract. [5] (2071 Magh)
13. Prepare bid invitation notice for the given project:
 Name of Project: - Construction of Municipal Building Ward no.4
 Office: - Dhaulagiri Municipality, Myagdi
 Estimated Amount: - Rs 1,80,00,000.00
 Method: - NCB, E-bidding
 Bid invitation date: - Today

[4] (2080 Chaitra)

- **5.4 Conditions of Contract**

- **5.5 Prequalification**

1. Explain about Pre-qualification of contractor. [4] (2079 Shrawan)
2. What is pre-qualification of contractor's and why it is necessary. [4] (2074 Bhadra)
3. Explain the pre-qualification and post-qualification of a contractor with their pros and cons. [Part of 5] (2074 Magh)
4. Write short notes on: Pre-qualification for the selection of contractors. [3] (2075 Bhadra)
5. Write short notes on: Prequalification of contractor. [4] (2070 Bhadra)
6. Write short notes on: Pre-qualification of contractors. [5] (2073 Magh)

- **5.6 Opening and Evaluation of Bids**

- **5.7 Award of Contract**

1. Explain how contract is awarded. [3] (2071 Magh)
2. Explain Performance security in brief. [2] (2080 Chaitra)

3. Calculate performance guarantee amount require for a contractor whose bidding amount is Rs 1,00,50,000. The estimated cost of the project is Rs 2,00,00,000. As per PPR 2064. [2] (2081 Shrawan)

- **5.8 Selection of Consultants**

1. Explain methods of selection of consultants as per PPA and PPR in Nepal. [6] (2081 Shrawan)
2. Explain about the process for evaluation of tender and selection of contractor. [8] (2069 Bhadra (Old Syllabus))
3. Write short notes on: Pre-bid meeting. [4] (2073 Bhadra)

- **5.9 Miscellaneous Terms**

1. Write short notes on: JV and sub-contracting. [3] (2079 Jestha)
-

Chapter 6: Construction Project Integration and Work [3 hours]

- **6.1 Site Surveying and Preparation**

1. List down the various components for site preparation. [3] (2075 Bhadra)
2. Explain the necessity of site surveying and preparation in construction project. List out components of site preparation. [3] (2072 Magh)

- **6.2 Arrangement of Facilities and Shops/Job Layout**

1. Explain the term Job Layout Plan. [3] (2078 Chaitra)
2. Define Job Layout plan with its example and figures. [3] (2080 Chaitra)
3. What is job layout plan? Explain with typical schematic diagram for a housing project. [5] (2079 Chaitra)
4. Prepare a typical job layout of a Building Project. [4] (2075 Bhadra)
5. Prepare a typical job layout of a Building Project. [4] (2071 Bhadra)
6. Prepare a typical job layout for a multi-story building construction project. [3] (2077 Chaitra)
7. Prepare a job layout plant for a Hospital Building Construction Site with construction site area of 500m by 500m. [4] (2081 Shrawan)

8. Prepare a job lay-out plan for constructing 1000 sqm area public building in a rectangular land ($190\text{ m} \times 125\text{ m}$) with sufficient facilities. [4] (2079 Jestha)
9. What is job layout? Discuss various factors to be considered in construction site planning. [2+6] (2069 Bhadra (Old Syllabus))
10. List down the various components for site preparation, and prepare a typical arrangement facilities of a building construction project in Urban area of Nepal. [4] (2073 Bhadra)
11. What are the different factors that need to be consider during construction site planning process? [4] (2078 Kartik)

• **6.3 Material Handling System**

1. Explain the concept of material handling. [3] (2072 Magh)
2. What is material handling system and why is it useful in construction work. [4] (2074 Bhadra)
3. Write short notes on: Material handling system. [4] (2073 Bhadra)
4. Write short notes on: Material handling system. [5] (2072 Ashwin)

• **6.4 Financial Management and Cash Flow Management**

1. Explain about the importance of financial and cash flow management for smooth running of construction project. [4] (2079 Shrawan)
2. Explain the points to be dealt in financial management and cash flow management by a manager in a construction project. [Part of 10] (2072 Ashwin)
3. What is cash flow management? [3] (2072 Magh)
4. What is cash flow and how do you prepare it? Explain its importance. [6] (2073 Magh)
5. Explain about the cash flow management in construction project. [3] (2075 Bhadra)
6. Discuss with examples cash-flow management during construction of project. [5] (2074 Bhadra)
7. How does mobilization advance affect the cash flow of contractor? [2] (2078 Chaitra)
8. Define financial management. Show a typical cash flow diagram of a contractor after contract agreement. [1+3] (2071 Magh)
9. Why cash flow management is necessary for a contractor. Draw a typical cash flow estimate of a contractor showing the most critical stage. [2+3] (2070 Magh)
10. Explain cash flow forecasting of a contractor depicting contract and contractor's margin with neat diagram. [3] (2077 Chaitra)
11. Write short notes on: Financial management. [4] (2070 Bhadra)

12. Write short notes on: Cash flow management in a construction project. [5] (2074 Magh)
 13. Write short notes on: Balance sheet. [4] (2069 Bhadra (Old Syllabus))
-

Chapter 7: Controlling Project Integration and Work [5 hours]

- **7.1 Work Scope Control**

1. Define work scope control. [2] (2071 Bhadra)
2. Explain in detail, what is project scope control and how it can be achieved? [4] (2077 Chaitra)
3. Explain the importance of work scope control in construction project. [Part of 6] (2073 Magh)
4. How do you control project scope from frequent changes? Explain. [4] (2081 Shrawan)

- **7.2 Product Quality Control**

1. Write short notes on: Product quality control. [2] (2072 Magh)

- **7.3 Labor Productivity Control**

1. Define Labor Productivity Index. What are the factors affecting labor productivity? Explain in brief. [6] (2080 Chaitra)
2. What are the major causes of Low labor productivity? [3] (2078 Chaitra)
3. What is labour productivity? What is to be done in order to improve labour productivity? [5] (2079 Shrawan)
4. How labor productivity can be measured? Write appropriate measures to increase labor productivity. [4] (2079 Jestha)
5. Define labour productivity. How can it be improved? Explain. [2+4] (2071 Magh)
6. Write short notes on: Labor productivity control. [5] (2074 Bhadra)
7. Write short notes on: Labor productivity control. [4] (2070 Bhadra)

- **7.4 Equipment Productivity Control**

1. Define equipment productivity control. [3] (2075 Bhadra)
2. What is equipment productivity? What are the causes of being low productivity and its remedial measures? [1+4] (2079 Chaitra)

3. Explain labor productivity and equipment productivity in construction industry. [5] (2076 Bhadra)
4. Explain labor productivity and equipment productivity. How can they improve? [5] (2074 Magh)

- **7.5 Material Productivity Control**

1. Explain about the material productivity control. [3] (2078 Chaitra)
2. Define material productivity and write down the basic reasons for material productivity control in construction project. [4] (2073 Bhadra)
3. Why it is necessary to do the material productivity control in the construction project? [3] (2075 Bhadra)
4. Explain the importance of material productivity control in construction project. [Part of 6] (2073 Magh)
5. Write short notes on: Material productivity control. [5] (2072 Ashwin, Q8)

- **7.6 Work Schedule Control**

➤ *No direct questions asked in past.*

- **7.7 Performance Control Using Earned Value Analysis**

- ❖ **Theory**

1. Discuss the Use of Earned value analysis for Performance control in a project. [6] (2075 Bhadra)
2. Discuss the use of earned value analysis's for performance control in a project. [4] (2074 Bhadra)
3. Discuss earned value analysis with example. [4] (2071 Bhadra)
4. Discuss how the performance control is carried out using earned value analysis explaining each terms used in earned value analysis. [4] (2072 Magh)
5. Explain how a manager takes decision in controlling project with "earned value analysis"? [Part of 10] (2072 Ashwin)
6. Explain the EVM technique for the cost control and monitoring. Illustrate an EVA graph with your own values. [5] (2076 Bhadra)
7. Write short notes on: Project cost control using EVA. [3] (2079 Jestha)
8. Write short notes on: Performance control using EVA. [4] (2070 Bhadra)

❖ Numerical

1. Perform EVA and comment on the performance for the given project

Activity	Cost (Rs.)	1	2	3	4	5	6	7
A	5000							
B	10000							
C	15000							
D	7000							
E	6000							
F	2000							

The progress was monitored and recorded at the end of 5th day. [7] (2078 Kartik)

Activity	% Completed	Expenditure (Rs)
A	100	5000
B	80	9000
C	60	8000
D	0	X
E	30	3500
F	0	X

2. Sinoma company of China has planned to prepare a Subgrade for Koteshwor to Kalanki road section of 8.7 km length of ring road to be accomplished within 1.5 years at a cost of 1.5 million per kilometer. The data were taken after 6 months of commencement of the work, the work found to be 2.3 km completed by expending 4.6 million rupees. Perform EVA and draw your conclusion regarding the Chinese performance in Nepalese domain by calculating each component of EVA. [6] (2077 Chaitra)
3. Perform earned value analysis of a project with five activities as given below. Progress was evaluated at the end of 5th day. [6] (2070 Magh, Q7)

Activity	Cost	1	2	3	4	5	6	7	8
A	6,000								
B	15,000								
C	10,000								
D	4,000								

At the end of 5 th day	% Complete	Actual Cost
A	100%	7000
B	55%	7500
C	40%	3600
D	0%	×

Also comment on the performances.

- Chino company of China has planned to prepare a Subgrade for Koteswor to Kalanki road of 8.7km length to be accomplished within 1.5 years at cost of Rs 1.5 million per kilometer. The data were taken after 6 month of the commencement of the work, the work found to be of only 2.3km completed by expending 4.6 million rupees. Perform EVA and draw your conclusion regarding the Chinese performance in Nepalese domain by calculating each component of EV. [5] (2074 Magh)

Chapter 8: Site Management [3 hours]

• 8.1 Responsibility of Site Engineer

- What are the roles and responsibilities of site engineer? Explain in detail. [4] (2077 Chaitra)
- What are the responsibilities of a site Engineer? [Part of 4] (2079 Shrawan)
- What are the responsibilities of site Engineer in construction site? [3] (2070 Magh)
- What are the responsibilities of site engineer in construction work? [4] (2069 Bhadra (Old Syllabus))
- Write the responsibilities of 'site engineer' while supervising the work of a contractor. [Part of 6] (2081 Shrawan)
- Write the responsibilities of 'site engineer' while supervising the work of a contractor. [Part of 10] (2072 Ashwin)
- What are the key responsibilities of a civil engineer in a hydropower project site? [Part of 5] (2074 Magh)
- Write short notes on: Responsibilities of Site Engineer. [3] (2070 Bhadra)

• 8.2 Supervisory Work of Contractor

- Explain in details about supervising work of contractor. [4] (2078 Kartik)

- **8.3 Record Keeping**

1. Define record Keeping and its importance in any Civil Construction work. [1] (2078 Kartik)
2. What are the vital record keeping documents for a public construction work? [2] (2079 Jestha)
3. Write short notes on: Importance of record keeping. [4] (2069 Bhadra (Old Syllabus))

- **8.4 Site Order Book**

1. What is site order book? [Part of 5] (2078 Chaitra)
2. What is site order book? [2] (2072 Magh)
3. Explain: Site Order Book [1] (2079 Jestha)
4. Explain site order book. [2] (2071 Magh)
5. Explain about the site order book. [3] (2075 Bhadra)
6. Prepare the format of Site order Book and Interim Payment Certificates. [4] (2080 Chaitra)

- **8.5 Procedures to Prepare Bills**

1. Write down the procedure of preparing bills. [Part of 5] (2078 Chaitra)
2. Explain procedures to prepare bills. [3] (2070 Magh)
3. Explain procedures to prepare Bills. [Not specified, assumed 8] (2071 Bhadra, Q8)
4. Differentiate between running bill and final bill. [2] (2077 Chaitra)
5. Differentiate between running bills and final bills. [2] (2072 Magh)

- **8.6 Measurement Book**

1. Explain: Measurement Book [1] (2079 Jestha)
2. Explain and differentiate between site order book and measurement book. [2] (2073 Bhadra)
3. Write short notes on: Measurement book. [Part of 5] (2073 Magh)
4. Write short notes on: Measurement book. [5] (2072 Ashwin)
5. Write short notes on: Measurement book. [3] (2070 Bhadra)

- **8.7 Muster Roll**

1. What is Muster Roll? [Part of 5] (2078 Chaitra)
2. Explain muster roll. [2] (2071 Magh)
3. Explain the terms: record keeping, site order book, muster roll, measurement book. [Part of 5] (2074 Magh)
4. What is muster roll and why it is important? [3] (2075 Bhadra)

5. What is muster roll and why it is important? [2] (2073 Bhadra)
 6. Prepare a typical muster roll used in construction project executed through user's committee. [2] (2077 Chaitra)
 7. Prepare a muster roll and the format of IPCs in a construction project. [2.5+2.5] (2079 Chaitra)
 8. Write short notes on: Muster Roll [Part of 5] (2073 Magh)
-

Chapter 9: Project Maintenance [4 hours]

• 9.1 Maintenance Basics

1. Explain project maintenance and its objective. [Part of 6] (2073 Magh)
2. Write down the objectives of the maintenance. [2] (2073 Bhadra)
3. Write the importance of project maintenance. [4] (2071 Magh)
4. Why maintenance is needed for a project? [Part of 4] (2078 Chaitra)
5. How do you see the importance of repair and maintenance to prolong the useful life of construction project? Explain. [2] (2077 Chaitra)
6. 'A Stitch in time saves nine'. Elaborate this statement with the objective of maintenance. [2] (2078 Kartik)
7. Write short notes on: Importance of Maintenance. [3] (2070 Bhadra)

• 9.2 Types of Maintenance

1. Describe the types of maintenance in a project. [4 (2071 Bhadra, Q4)]
2. Describe different types of maintenance in a project and explain the importance of maintenance in operating project. [5] (2079 Chaitra)
3. Describe the types of maintenance in a project? Why maintenance is necessary in construction project? [5] (2079 Shrawan)
4. Describe the types of maintenance in a project and explain the importance of maintenance of already built projects. [6] (2074 Bhadra)
5. Differentiate different types of project maintenance. [Part of 6] (2073 Magh)
6. Explain about planned maintenance. [4] (2078 Chaitra)
7. Differentiate between planned maintenance and unplanned maintenance with examples. [2] (2079 Jestha)

8. Differentiate between planned maintenance and unplanned maintenance with suitable examples. [2] (2076 Bhadra)
9. Define preventive and corrective maintenance. [2] (2077 Chaitra)
10. Explain preventive, corrective and breakdown maintenance with examples. [3] (2078 Kartik)
11. Explain and differentiate between preventive and corrective maintenance giving suitable examples. [4] (2072 Magh)
12. Enunciate different types of maintenance. Discuss "preventive maintenance is better than corrective maintenance". [5] (2074 Magh)
13. Justify the statement 'prevention maintenance is better than corrective maintenance.' [Part of 6] (2081 Shrawan)
14. Justify the statement 'prevention maintenance is better than corrective maintenance'. [Part of 10] (2072 Ashwin)
15. "Prevention is better than cure" Elaborate it with reference to preventive and corrective maintenance. [4] (2070 Magh)
16. "Preventive maintenance is better than the corrective maintenance". Do you agree with this statement? Explain with suitable example. [8] (2069 Bhadra (Old Syllabus))
17. Write short notes on: Types of maintenance in Buildings. [3] (2077 Chaitra)
18. Write short notes on: Diagnostic Maintenance. [3] (2070 Bhadra)

- **9.3 Planning and Scheduling of Maintenance**

1. Explain the term planning and scheduling of maintenance. [Part of 4] (2072 Magh)
2. Explain in details about planning and scheduling of maintenance in project maintenance. [3] (2073 Bhadra)
3. How a good maintenance plan can be formulated, explain in brief. [2] (2079 Jestha)
4. How a good maintenance plan can be formulated, explain in brief. [2] (2076 Bhadra)

- **9.4 Estimating Maintenance Cost**

➤ *No direct questions asked in the past.*

- **9.5 Management of Maintenance and Financing**

➤ *No direct questions asked in the past.*

Chapter 10: Personnel Management [4 hours]

- **10.1 Management Principles: Administration and Organization Principles**

1. Define administration and organization. Briefly explain any two types of organization. [2+4] (2071 Magh)
2. Write short notes on: Fayol's principles of management. [4] (2069 Bhadra (Old Syllabus))
3. List out the 14 administrative management principles propounded by Fayol and elaborate any six of them. [6] (2079 Chaitra)
4. Write short notes on Administrative Principles. [5] (2079 Shrawan)

- **10.2 Centralization and Decentralization**

1. Write short notes on: Advantage and disadvantage of centralization. [5] (2074 Bhadra)
2. Explain the benefits of decentralization in Project Control and Management. [4] (2080 Chaitra)
3. What are different techniques by which administrative and market decentralization can be achieved? [2] (2079 Jestha)

- **10.3 Supervisory and Leadership Styles**

1. What is leadership? What are the characteristics of a good leader? [Part of 6] (2078 Chaitra)
2. Define leadership. What are the characteristics of good leader? Which style is better? [1+3+2] (2071 Bhadra)
3. Explain different types of leadership styles. [4] (2073 Magh)
4. Discuss leadership styles. [3] (2072 Magh)
5. Explain different types of leadership style with suitable examples. [2] (2079 Jestha)
6. Explain various types of leadership and their suitability in construction projects in Nepal. [6] (2070 Magh)
7. Write short notes on: Leadership styles. [3] (2075 Bhadra)
8. Write short notes on: Leadership styles. [5] (2072 Ashwin)
9. Write short notes on: Leadership Styles. [4] (2070 Bhadra)

- **10.4 Information System for Decisions**

1. How an effective information system helps for quality decision making process, explain it.
[Part of 4] (2072 Magh)

- **10.5 Motivation: System for Decisions, Human Elements, Evaluation and Merit Rating, and Directing**

1. Why motivation for workers is important in any construction projects? [Part of 6] (2078 Chaitra)
2. Why motivation is necessary for productive output from people? Explain Maslow's Hierarchy of Needs. [6] (2081 Shrawan)
3. How personnel involved in any organization can be motivated? Explain at least two theories of motivation. [2+4] (2076 Bhadra)
4. Describe Herzberg Dual factor theory of motivation. [1] (2077 Chaitra)
5. Write short notes on: Maslow's theory of motivation. [4] (2069 Bhadra (Old Syllabus))
6. Write short notes on : Herzberg Dual factor theory of motivation [5] (2076 Bhadra)
7. Write short notes on: Motivating and directing. [4] (2070 Bhadra)

- **10.6 Personnel Selection, Testing, and Training**

1. Define personal management. [Part of 4] (2072 Magh)
2. Define personnel management. Write down the steps of personnel selection in any organization. [1+4] (2078 Kartik)
3. Define personnel management. Write down the steps of personnel selection in any organisation. [2+4] (2073 Bhadra)
4. Write down the steps of personal selection in any organization. [3] (2077 Chaitra)
5. Write down the steps of personal selection in any organization. [3] (2072 Magh)

- **10.7 Trade Unions and Relation with Management**

➤ *No direct questions asked in the past.*

- **10.8 Effective and Timely Communication**

1. What is communication in a project? Discuss its importance. [4] (2074 Bhadra)
2. Explain the importance of communication in construction project. [Part of 10] (2072 Ashwin)
3. "Effective and timely communication is like blood in any organization". Elaborate on the ground of personnel management. [5] (2074 Magh)
4. Write short notes on: Importance of communication. [5] (2073 Magh)

5. Write short notes on: Barriers to Communication. [3] (2077 Chaitra)
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Chapter 11: Regulatory Requirements

- **11.1 Safety Requirements**

1. Why regulatory requirements are needed in any construction project? [Part of 6] (2080 Chaitra)
2. Why regulatory requirements are needed in any construction project? [Part of 4] (2073 Bhadra)
3. Explain in brief the physical and physiological causes of accident in construction project. [3] (2081 Shrawan)
4. What are the causes of accidents in construction project? How it can be minimized? [2+2] (2071 Bhadra)
5. Discuss about safety requirements in a construction site. [4] (2073 Bhadra)
6. Why construction safety is important in any construction project? [2] (2079 Jestha)
7. Why construction safety is important in any construction project? What are the different causes of accidents in the construction site? Explain in brief. [2+2] (2078 Kartik)
8. Discuss about the importance of safety requirements in the construction site. [2] (2075 Bhadra)
9. Discuss various safety control devices used in construction site to ensure safety. [5] (2078 Chaitra)
10. Write short notes on: Safety Requirements. [3] (2070 Bhadra)

- **11.2 Workman's Compensation Board**

➤ *No direct questions asked in the past.*

- **11.3 Fire Regulations and Insurance**

1. Write short notes on Fire Regulations and safety Requirements in site [5] (2079 Shrawan)
2. Why insurance is required in a construction project? Explain with the headings that are insurable in a construction project. [2+4] (2079 Chaitra)
3. Why insurance is necessary during construction? Explain. [4] (2070 Magh)

- **11.4 Environment Concern and Protection**

1. Discuss about environmental concerns in project. [4] (2071 Magh)

- **11.5 Building Codes and Quality Control**

1. Which National Building Codes to be perused in Nepal for construction safety and safety regulations. [1] (2081 Shrawan)
 2. Explain NBC hierarchy and its current implementation status to regulate the building construction work in the country. [2] (2079 Jestha)
 3. How building codes and bye laws of any country helps in effective urban planning? Explain it. [Part of 6] (2080 Chaitra)
 4. How building codes of any country helps in effective urban planning? [3] (2075 Bhadra)
 5. How building codes of any country helps in effective urban planning, explain it. [Part of 4] (2073 Bhadra)
 6. Write short notes on: National building code. [5] (2074 Bhadra)
 7. Write short notes on: Building codes. [3] (2070 Bhadra)
 8. Write short notes on: Building Bye-Laws and its requirements. [3] (2077 Chaitra)
 9. Write short notes on: Use of Building code and Quality Control. [5] (2073 Magh)
 10. Write short notes on: Building codes and quality control [2] (2072 Magh)
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Chapter 12: Specification [6 hours]

- **12.1 Purpose of Specifications**

1. What are purpose of specification? [Part of 7] (2079 Shrawan)
2. Write the purposes of specification writing. [Part of 4]
3. State the purpose of specification. [Part of 5] (2072 Ashwin)

- **12.2 Types of Specifications: General and Detailed Specifications**

1. Define specification and explain its types. [4] (2078 Chaitra)
2. Enunciate different types of specification. [1] (2079 Jestha)
3. Enunciate different types of specification. [2] (2078 Kartik)
4. Discuss the types of specification. [Part of 5] (2074 Magh)
5. What are different types of specification you are aware of? [Part of 6] (2074 Bhadra)

6. Explain detailed specification. [4] (2071 Magh)
7. Define and differentiate between general and detailed specification. [2] (2077 Chaitra)
8. Differentiate between general and detailed specification. [2] (2070 Magh)

- **12.3 Importance of Specifications**

1. Explain the importance of specification in construction work. [4] (2069 Bhadra (Old Syllabus))
2. Explain the importance of Specifications in Construction Works. [4] (2070 Bhadra)
3. Why specification is important in construction contracts? [2] (2075 Bhadra)
4. Why specification is important in construction contracts? [2] (2073 Bhadra)
5. Why specification is important in road construction project? Explain. [2] (2078 Kartik)
6. Write the importance of specification. [Part of 6] (2074 Bhadra)
7. Explain the importance of specification. [3] (2071 Bhadra)
8. Discuss the importance of specification in construction work. [Part of 6] (2073 Magh)
9. Explain the roles of specifications in any construction project. [3] (2079 Chaitra)
10. What are the importance of specification and discuss different elements used in writing a detailed specification? [Part of 5] (2074 Magh)

- **12.4 Specification Writing: Technique, Use of International and Local Standards, Codes of Practice**

1. What are the principles of writing specification? [1] (2079 Jestha)
2. What are the principles of writing the specification? [4] (2078 Kartik)
3. What are the principles of writing technical specification for construction works? [3] (2079 Chaitra)
4. Discuss on techniques of writing specification. [4] (2078 Chaitra)
5. Write the techniques of specification writing. [Part of 5] (2072 Ashwin)
6. Write down the major techniques that you follow for writing specification. [3] (2075 Bhadra)
7. Write down the main techniques that you follow for writing specification. [4] (2069 Bhadra (Old Syllabus))
8. Write a note on the art and science of writing Specifications covering following aspects: (i) Necessity of specification (ii) Procedure of drafting (iii) What to write and what to avoid in writing. [6] (2070 Bhadra)
9. Explain with examples why international standards are used in drafting of specification. [2] (2072 Magh)

- **12.5 Typical Examples of Specifications**

1. Write down detailed specification for cement sand mortar 1:6. [Part of 6] (2073 Magh)
 2. Write a detailed specification of brick work in cement sand mortar. [5] (2071 Bhadra)
 3. Write detailed specification for Brick work in 1:4 cement sand mortar. [7] (2079 Shrawan)
 4. Write down the specification paragraphs for first class brick works in cement mortar of 1:4. [4] (2072 Magh)
 5. Write detailed specification for brick work in 1:4 cement sand mortar. [7] (2070 Magh)
 6. Write detailed specification of First Class Brick work with (1:6) cement sand mortar. [2+6] (2080 Chaitra)
 7. Write detailed specification of M20 RCC works.
 8. Write detailed specification of: (i) RCC with M20 concrete in column (ii) Brick work in 1:6 cement sand mortar. [4+4] (2069 Bhadra (Old Syllabus))
 9. Write down the specification for Stone masonry work in 1:6 mortar. [4] (2073 Bhadra)
 10. Write a detailed specification of cement plaster of 1:4 in brick wall. [4] (2077 Chaitra)
 11. Prepare detailed specification for earth work in excavation in sandy soil for isolated footing of multi-storey building. [6] (2079 Jestha)
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Chapter 13: Valuation [7 hours]

- **13.1 Introduction**

1. What is valuation? [Part of 6] (2074 Bhadra)
2. Define valuation. [2] (2070 Magh)

- **13.2 Cost and Value**

1. Define cost and value. [Part of 8] (2076 Bhadra)
2. Differentiate between cost and value. [Part of 6] (2073 Magh)
3. Differentiate between cost and value. [Part of 8] (2071 Magh)
4. Write short notes on: Scrap Value and Salvage value. [3] (2075 Bhadra)

- **13.3 Purpose of Valuation and Principle of Valuation**

1. Why valuation is done? [2] (2078 Chaitra)

2. Why valuation is done? [3] (2076 Bhadra)
3. What are the purposes of valuation? [3] (2070 Magh)
4. Define value of any property and what is to be satisfied by any property to have its value. [1] (2073 Bhadra)
5. Explain importance of specification and valuation. [Part of 5] (2074 Magh)

- **13.4 Factors Affecting the Value of the Property**

1. What are the various factors affecting the value of the property. [2] (2077 Chaitra)
2. Explain different factors affecting valuation of property. [Part of 6] (2073 Magh)
3. Explain the factors affecting value of the property. [3] (2070 Magh)
4. Discuss with examples about the factors affecting the value of the property. [2] (2072 Magh)
5. Explain the different parameters affecting the valuation of the property. [2] (2078 Chaitra)
6. Explain different parameters affecting the valuation of the property. [3] (2076 Bhadra)

- **13.5 Value Classification**

1. Explain value classification. [Part of 8] (2071 Magh)

- **13.6 Sinking Fund**

1. The annual sinking fund of concrete vibrating machine costing Rs.50,000 is Rs. 150 and its salvage value is estimated to be Rs.5,000 only. Assuming interest rate as 4%. Determine the useful life of the machine. [4] (2072 Magh)

- **13.7 Capitalized Value**

- (Covered in various valuation problems below.)

- **13.8 Obsolescence**

1. Define obsolescence. [Part of 8] (2076 Bhadra)

- **13.9 Depreciation**

1. Elaborate concept of depreciations. [Part of 5] (2072 Ashwin)

- **13.10 Qualification of a Valuer**

1. Explain on qualification of valuer. [Part of 8] (2076 Bhadra, Q3b)

- **13.11 Valuation of Land**

- (Covered in various valuation problems below.)

- **13.12 Various Methods of Valuation of Properties**

1. Write different methods of valuation. [Part of 6] (2074 Bhadra)
2. Write various methods of valuation of properties. [Part of 5] (2072 Ashwin,)
3. What are the different methods of working out the valuation of property? [Part of 5] (2074 Magh)
4. Explain the situation where the following method of valuation will be used: (i) Cost based method (ii) Plinth area method (iii) Development method. [3] (2070 Bhadra)
5. Write short notes on: Methods of valuation of properties. [4] (2073 Bhadra)

- **13.13 Role of Computers in Valuation**

- *No direct questions asked in the past.*

- **13.14 Report Writing**

- *No direct questions asked in the past.*

- ❖ **Valuation Numericals**

1. A prospective investor desires to purchase a piece of converted land measuring 1,56,000 sqft in area. After developing this land for housing colony, he wants to sell the plots (1700 sqft in area) at the rate of Rs 9300 per sqft which is prevailing in the vicinity and thus earn net profit of 22% in the bargain. With the following information, work out the valuation of the undeveloped property. Express the value of property per sqft basis. [7] (2081 Shrawan)
 - a) Deduction for roads, parks, electric sub-stations = 30% area
 - b) Levelling, dressing and jungle clearance cost = Rs 5 per sqft
 - c) Cost of providing roads, drainage, w/s and electric sub-station = Rs 400 per sqft
 - d) Engineer's fee and miscellaneous charges = 5% of sale price
 - e) Legal charges, brokerage and advertisement = 10% on sale price
2. A freehold property consisting of RCC building having a total area of 500 m² constructed 10 years ago is proposed to be purchased by Mr. Thapa. The present market rate of construction for similar type of building is Rs 45,000/m². The total area of land is 500 m². The commercial market rate (CMR) of this land is Rs 1,00,000/m². The government registration rate (GRR)

for this land is Rs 30,000 /m². Use 60:40 proportion for determining cost of land. Assume life of building to be 70 years and scrap value as 10%. Advise Mr. Thapa as to what cost he should purchase the property. Use Plinth area method of valuation and straight line method of depreciation. [6] (2080 Chaitra)

3. A building having a cubic of 1600 m³ was constructed 14 years ago on a free hold land measuring 900 m². The building fetched a rent of Rs. 12300 per month. What amount will you recommend for advancing a loan on a property against the Mortgage, if the rate of land in that area is Rs. 800 per sq.meter? The building has following expenditure headings:

Insurance premium Rs. 6700 per annum

Repair and maintenance 7% of gross rent taxes 27% of gross rent

Management collection charges 5% of gross rent [6] (2079 Chaitra)

4. A three storied RCC building with expected 65 years of useful life is built on a plot of land measuring 550 m². The plinth area of each story is 325 m² and it fetches on a gross rent of 12,000 per month. Calculate the capitalized value of property based on 7.5% net yield. The sinking fund is assumed to be computed in 3% interest rate. The cost of land is Rs.415 per m². The plinth area rate of similar building in the vicinity was found to be Rs. 22,500 per m². [7] (2079 Shrawan)

5. Assess fair market value of Tata Sipradi Showroom with following information

Covered land area = 14 aana @ Rs 95,00,000 per aana prevailing rate

Built up Showroom = 3355 sqft @ Rs 7200/sqft of current construction

Gross monthly income of Showroom = Rs 80,00,000

Municipal Tax = 8% of gross income

Monthly Staff Salary and administrative expenses = Rs 2,90,000

Advertisement expenses (annual) = Rs 60,00,000

Lifting and other installed equipment costing Rs 58,00,000 with estimated life of 15 yrs undertaken for salvage value with 30% estimated salvage at the end with 7% interest per annum

Electricity, customer courtesy, solid waste disposal, drinking water supply and other miscellaneous cost = 15% of gross income

Insurance premium = Rs 40,00,000

Maintenance cost = Rs 50,000 per month

Consider year's purchase for 60 yrs with prevailing interest rate 8% and 10% redemption of capital. [2+6] (2079 Jestha)

6. A building, having estimated future life of 80 years, fetches a gross annual rent of Rs. 2500 per month. Work out its capitalized value on the basis of 8% net yield. The rate of above building measures 1200 m² and cost of land is taken at Rs. 100 per m².

The other outgoing are:

Repair and maintenance (1/12)th of gross income

Municipal axis and property tax – 25% of gross income

Management and miscellaneous charges – 8% of gross income

The plinth area of a building is 600 m² and plinth area rate of the building is Rs. 200 per m².

[6] (2078 Chaitra)

7. A building fetches a gross rent of Rs. 20,000.00 per month. Assuming 25% outgoings for taxes, 20% for repairs and maintenance and 5% to set aside to accumulate the cost of building. Determine its capitalized value at the rate of interest on capital as 6% per annum and 4% on sinking fund installment. [5] (2078 Kartik)
8. Work out the valuation of a cinema hall with the following [6] (2077 Chaitra)
- a) Cost of land = Rs. 15,00,000
 - b) Gross income = Rs. 90,00,000
 - c) Expenses undergone per year to run the Cinema including staff salary, electricity charges, municipal taxed, stationary and printing, etc. 30% of gross income
 - d) Repair and maintenance of machinery, plant and equipment at 5% of their capital cost which is Rs. 55,00,000
 - e) Sinking fund for machinery, plant and equipment whose life is estimated at 25 years at 5% after allowing 10% scrap value.
 - f) Insurance premium is 60,000 per annum
- Assume year's purchase for 60 years at 8% and redemption of capital at 4% and repair of hall at 2% of gross income.
9. An office building having a plinth area of 4000 sq ft. was constructed at the cost of Rs. 4000.00 per sq. ft. The building is two storey RCC structure on a 4000 sq. ft. plot. The cost of land is Rs. 1000000.00. The office building is given on a monthly rental of Rs. 50000.00. Life of the structure may be taken as 70 years. For sinking fund 6% compound interest is prescribed. Work out the capitalized value of the property on the basis of net yield of 8%. Repairs and maintenance charges may be taken as one month rent and municipal and property

tax 25% of annual rentable value. Insurance and other miscellaneous charges is 8% of annual rentable value. [10] (2076 Bhadra)

10. Calculate the value of a building built 20 years ago having plinth area 500 sqm, constructed in the land of 1000 sqm. Current market value of the land is Rs. 25000 per sqm and rate of building is Rs. 50,000 per sqm. Assume suitable data if necessary. [6] (2075 Bhadra)
11. A RCC framed structure building having estimated future life 80 years, fetches a gross annual rent of Rs 2200 per month. Work out its capitalized value on the basis of 6% net yield. The rate of compound interest for sinking fund may be taken 4%. The land plot of above building measures 1400 sqm and cost of land may be taken to be Rs.120 per sqm. The other outgoing are: (i) Repair and maintenance $\frac{1}{12}$ th of the gross income (ii) Municipal taxes and Property tax 25% of gross income (iii) Management and Miscellaneous charges 7% of gross income [5] (2074 Magh)
12. A building having a cubic of 1500 m³ was constructed 15 years ago on a free hold land measuring 1000 m². The building fetched a rent of Rs.12000 per month. What amount will you recommend for advancing a loan on a property against the Mortgage, if the rate if land in that area is Rs.700 per sq meter? The building has following expenditure headings:
Insurance premium Rs. 6000 per annum
Repair and maintenance 8% of gross rent
Taxes 25% of gross rent
Management collection charges 5% of gross rent
Redemption of capital is 5%

[6] (2073 Magh)
13. A person desire to sale his property having details as: Gross rent = Rs 5000 per month, total outgoings = 25% of gross rent, estimated future life = 35 yrs, area of land = 200 m², estimated value of Land = Rs 2000 per square meter. Determine the fare value of property. Take rate of interest on capital as 10% and rate of redemption of capital as 5%. Assume suitable data, if required. [6] (2073 Bhadra)
14. Workout the value of a super market from the following data. Assume necessary data. [8] (2071 Magh)
 - a) Cost of land = Rs. 10 crore
 - b) Gross income = Rs. 5 crore
 - c) Repair and maintenance of machine = Rs. 5 lakh
 - d) Insurance premium = Rs. 100,000
 - e) Years purchase for 50 years at 9% and redemption of capital at 10%

15. Calculate the value of a building built 10 years ago having a plinth area of 450 sqm, constructed in the land of 1000 sqm. Current market value of land is Rs. 15,000 per sqm and rate of building is Rs. 30,000 per sqm. Assume suitable data if necessary. [8] (2071 Bhadra)
16. Workout the value of a Cinema hall from the given data. [8] (2070 Magh)
- i) Cost of Land = Rs 5 crore
 - ii) Gross Income = Rs. 2 crore
 - iii) Operating Cost = 40% of Gross income
 - iv) Repair and Maintenance of machineries = 5% of Capital Cost where capital cost is Rs. 40 lakh
 - v) Repair of Hall = 5% of Gross income
 - vi) Sinking fund for machineries whose life is estimated 25 years @ 4% after allowing 10% scrap value.
 - vii) Insurance premium = Rs 50000 per year
 - viii) Assume years purchase for 60 years at 8% and redemption of capital at 10%.
17. Work out the valuation of a Cinema hall with the following data. [5] (2070 Bhadra)
- (i) Cost of land = Rs. 15,00,000
 - (ii) Gross income = Rs. 90,00,000
 - (iii) Expenses undergone per year to run the Cinema including staff salary, electricity charges, municipal taxes stationary and printing etc. 30% of gross income.
 - (iv) Repair and maintenance of machinery, plant and equipment at 5% of their capital cost which is Rs. 55,00,000
 - (v) Sinking fund for machinery, plant and equipment whose life is estimated as 25 years at 5 % after allowing 10% scrap value
 - (vi) Insurance premium is Rs 60,000 per annum
- Assume year's purchase for 60 years at 8% and redemption of capital at 4% and repair of hall at 2% of gross income.